A cellular digital twin of *Arabidopsis* germination under spaceflight hardware environments

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Abstract

Seeds are the starting point of every crop grown off Earth, yet spaceflight experiments sample germinating seeds rarely, and the environment a seed experiences inside flight hardware is seldom measured. We built an open, browser-based digital twin of an *Arabidopsis thaliana* seed that follows one seed from imbibition to a five-day seedling. It couples a cellular embryo template indexed as in Bassel et al. [1], seed-level water uptake, O_2, CO_2 and ethylene exchange through the boundary layer, water film, testa and endosperm, the ABA–GA–Integrator network of Abley et al. [2] in every cell, Lockhart growth under seed-coat restraint, and seedling kinematics. Environments are generated from curated records of flight hardware (BRIC, CARA, APH, EMCS, LEAF GBC1), NASA exploration atmospheres and measured radiation dose rates. Of 164 parameters, 8 are measured, 33 standard, 22 derived, 15 fitted and 86 assumed; each is listed with its source. Calibrated to published germination timing, the twin reproduces testa rupture at 24.0 h, endosperm rupture at 32.0 h and peak relative expansion at cortical cell 7. With between-seed variability taken from a measured hydrotime spread, a simulated population germinates with t_{50} = 33.3 h and a 16–84% span of 20.9 h — about twice the spread that hydrotime theory gives for the same seeds (10.1 h); the twin is also more sensitive to low water potential than hydrotime predicts. Across hardware, the twin predicts that seed-coat and tissue resistance, not headspace depletion, sets embryo O_2, and that gravity acts on the seed mainly through the water film and seedling tropisms. These outputs are hypotheses to be tested against the companion DeepSpace Seed Stress Decoder and flight data.

1 Introduction

Germination converts a dry, metabolically quiescent seed into a growing seedling within about two days. In *Arabidopsis* it is completed by expansion of cells in the lower hypocotyl and transition zone rather than the radicle [3]. Bassel et al. [1] showed with a 3D cellular finite-element model that the size, shape and arrangement of embryo cells displace this growth away from the centre of growth-promoting gene expression; their model did not include the mechanical restraint of the seed coat and endosperm, nor any environment.

Spaceflight hardware changes exactly the variables that model leaves out: gas exchange, water distribution, ethylene accumulation and radiation. The companion DeepSpace Seed Stress Decoder maps which seed cell types carry the transcriptional signature of each stressor. Here we build the mechanistic counterpart — a time-resolved, cellular digital twin — to ask how a given hardware environment reaches those cells.

2 Methods

2.1 Parameter ledger

Every parameter is a row of `data/params/parameter_ledger.csv` with value, unit, source and evidence tier (measured, standard, derived, fitted, assumed); the build refuses a non-assumed row

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without a resolvable source. References were rebuilt from Crossref metadata. Source searches, including what could not be found, are archived in `data/params/research`.

2.2 Geometry

No open segmented 3D mesh of the mature *Arabidopsis* embryo was found (3DCellAtlas data are available on request; the MorphoGraphX2 deposit contains no embryo). The template is therefore procedural and schematic. The seed is an ellipsoid of 490 μm length [4]; the embryonic axis has 25 slices (7 radicle cells below the collet, which Bassel et al. place at cortical cell 8, and 18 hypocotyl cells per file [5]), with ring cell numbers derived from the 3DCellAtlas counts of the embryo axis [6] and two hypocotyl cortex layers [7]. Each cell is assigned a germinating-embryo cluster of the single-cell atlas of Liew et al. [8] so that decoder results can be mapped onto it.

2.3 Water, gas exchange and energy

Water uptake follows first-order imbibition to a plateau that declines with substrate water potential. Turgor above the wall-yield turgor of a reference collet cell scales with $\psi - \psi_b$, so that growth of that cell stops at the hydrotime base water potential ψ_b [9]. O_2, CO_2 and ethylene pass in series through the air boundary layer, a water film, the testa and the endosperm. The air-side coefficient is $\mathrm{Sh} = 2 + 0.6\,\mathrm{Re}^{1/2}\mathrm{Sc}^{1/3}$ [10], which reduces to $\mathrm{Sh} = 2$ in still air; gas-phase diffusivities scale as $1/P$ [11, 12]; Henry solubilities are from [13]. The wetted fraction and film thickness interpolate between 0 g and 1 g values with a dish-scale Bond number. Inside the embryo, O_2 follows the quasi-steady profile of a cylinder (axis) or slab (cotyledons) with uptake, using a tissue diffusivity from *Brassica napus* seed tissue [14] as a proxy. Seed O_2 uptake at full activity is set from a developing-seed respiration rate [15] and seed mass [4]. The headspace of each hardware unit is a well-mixed volume shared by its seeds, with ventilation, leak and scrubbing integrated exactly.

2.4 Hormones

Each cell carries the ABA–GA–Integrator model of Ableley et al. [2] with its published equations and default parameters, in the published monostable regime appropriate to after-ripened seeds. The twin adds: the sowing-induced GA signal graded from the radicle tip and a GID1 gradient, as specified by Bassel et al. [1]; ABA/GA exchange between neighbouring cells at the transport rate used by Topham et al. [16]; imbibition-induced ABA catabolism, the CYP707A2 mechanism of Okamoto et al. [17]; and ethylene, made from ACC by ACC oxidase with an O_2 requirement [18, 19], which lowers the Integrator [20]. The ethylene response uses the dose–response of [20].

2.5 Growth and restraint

Each cell grows by Lockhart's law with a wall stress $\sigma = P\rho/2t$ that rises with cell size, using the wall thicknesses and turgor of Bassel et al. [1]; cells of a slice share one axial strain rate weighted by wall cross-section. The new element is the seed coat: the testa and then the micropylar endosperm resist axial growth until their strength is exceeded; the endosperm cap weakens in proportion to the embryo GA signal and ethylene, held back by ABA.

2.6 Seedling, radiation and environments

After protrusion, roots follow measured kinematics [21], with gravitropic bending scaled by gravity; hypocotyl cells elongate without division up to lengths derived from meta-analysed final lengths [22]. Ethylene shortens the root and hypocotyl, thickens the hypocotyl and exaggerates the hook. Radiation scenarios use the ISS dose measured by the Environmental Data App, lunar-surface [23], cruise [24] and solar-particle-event estimates [25]; links from radiation damage to hormones default to zero because no verified source quantifies them. Hardware presets are generated from curated,

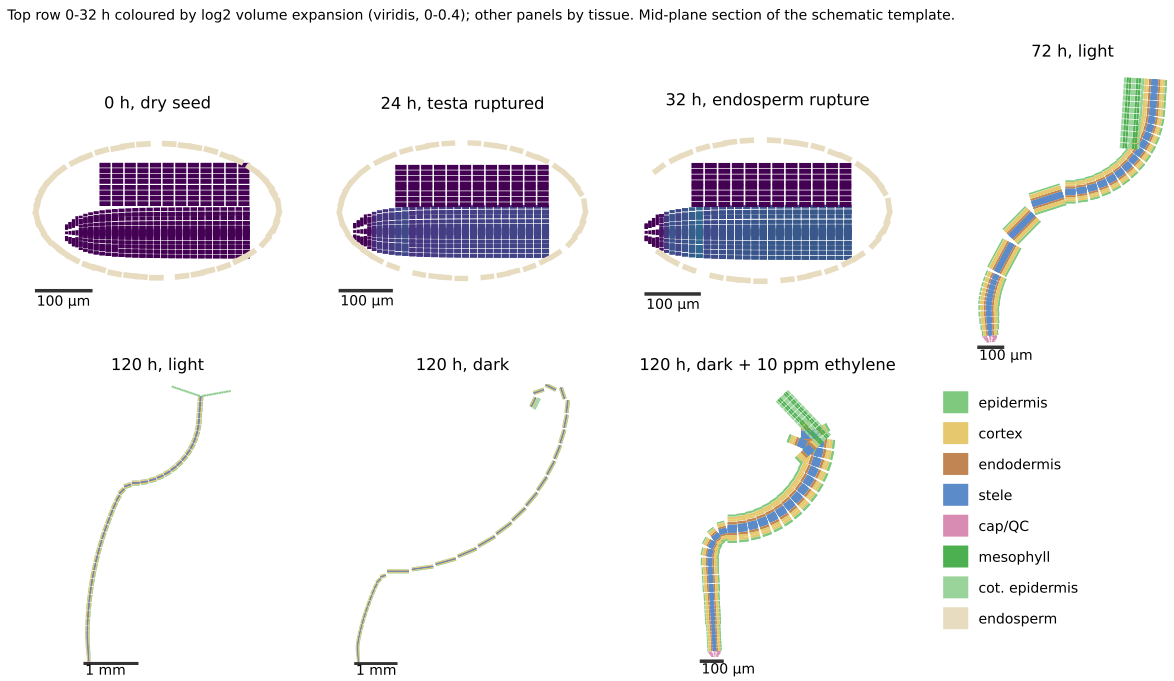


Figure 1: The twin through time: mid-plane sections of the schematic cellular seed. Germination panels are coloured by cell volume expansion; seedling panels by tissue. At 120 h, compare light, darkness and darkness with 10 ppm ethylene (triple response).

quoted records. For LEAF GBC1, a two-dimensional lattice-Boltzmann run of the chamber cross-section with the validated LunarLeaf-CFD solver gives an air speed at the seed bed of 0.053 mm s^{-1} at 0.25 L min^{-1} (outflow/inflow 0.95).

2.7 Calibration and validation

9 germination parameters were fitted by Nelder–Mead to 6 observed targets (loss 0.44), and 6 seedling parameters to seedling targets (loss 4.29). Because the number of fitted parameters exceeds or matches the number of targets, agreement shows that the model can reproduce the targets, not that it is identified. We therefore report out-of-sample checks separately. 26 validation gates (26 passing; analytic diffusion profile, still-air Sherwood limit, $1/P$ scaling, headspace mass balance, no growth below yield, no rupture under infinite restraint, observed timing windows, triple response, gravitropism, provenance) run in continuous integration. Numerical kernels were re-implemented independently in Python and agree with the TypeScript to 10^{-9}.

3 Results

3.1 Calibrated germination

The seed, its cells and their growth are shown through time in Fig. 1. The calibrated seed ruptures its testa at 24.0 h and its endosperm at 32.0 h (observed: 20–31 h and 31–45 h for Col-0 [26]; 28–32 h for the endosperm-rupture phase [1]). Lower-hypocotyl expansion begins at 19.0 h (observed 16–22 h) and relative expansion peaks at cortical cell 7 (observed ≈8) (Fig. 2). ABA falls to 0.30 of its initial level by 12 h (observed 0.30 [17]), and the endosperm cap ruptures at 0.60 of its initial strength (observed ≈0.6 in *Lepidium* [27]). In seedlings, cotyledons emerge at 65 h and open at 73 h (observed 60 and 72 h [28]). Peak hypocotyl growth is 5.99 and 1.22 mm d^{-1} in darkness and light (observed 5.96 and 1.11 [22]). Seedling ethylene emission is 0.40 pL h^{-1} (observed 0.26–0.53 [29]).

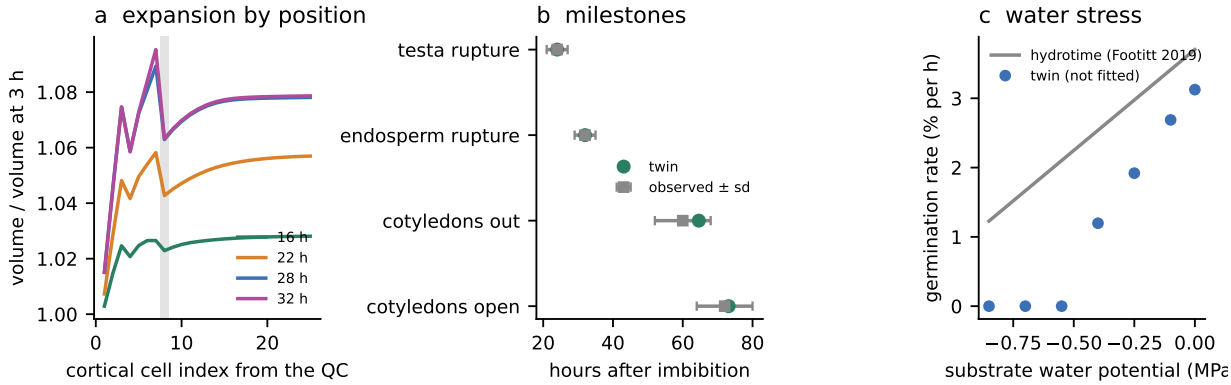


Figure 2: Calibration and an out-of-sample test. (a) Cortical cell volume relative to 3 h by position from the quiescent centre; grey: observed peak position (Bassel et al. 2014). (b) Simulated milestones against observed values. (c) Germination rate against substrate water potential: twin (points, not fitted to these data) and the hydrotime model of Footitt et al. 2019 (line).

Root elongation at 120 h, 88 μm h^{-1}, falls below the rate extrapolated from day-6 kinematics, a discrepancy we report rather than fit away.

3.2 Out-of-sample checks

Neither check below was used in calibration.

Population spread. We drew each seed's base water potential from the measured between-seed spread (σ_{ψ_b} = 0.222 MPa [9]). 98% of 60 seeds then rupture the endosperm, with t_{50} = 33.3 h and a 16–84% span of 20.9 h. For the same seeds, hydrotime theory gives t_{50} = 27.8 h and a span of 10.1 h, and Col-0 seed lots complete endosperm rupture between 31 and 45 h [26]. The twin therefore turns a given spread of ψ_b into a wider spread of germination times than hydrotime does.

Water stress. Lowering the substrate water potential slows the twin's mean seed considerably more than hydrotime predicts:

	0 MPa	−0.25 MPa	−0.4 MPa
Twin (h)	32.0	52.1	83.5
Hydrotime (h)	27.0	33.6	39.4

At −0.55 MPa and below the twin does not germinate at all (none by 200 h), whereas hydrotime predicts 47.6 h at −0.55 MPa and 81.4 h at −0.85 MPa (at 15 °C). The twin's effective base water potential is thus near −0.5 MPa rather than −1.27 MPa. The most likely cause is the *assumed* decline of the imbibition plateau with water potential: it also scales metabolic activation and the GA signal, adding to the hydrotime-linked turgor. We report this mismatch rather than refit that parameter, which would turn the check into an in-sample fit.

3.3 Hardware environments

In every hardware preset, embryo-surface O_2 falls well below headspace O_2 (Fig. 3). Headspace depletion is smaller: even in sealed BRIC-PDFU, headspace O_2 falls from 21.2 to 18.4 kPa over 120 h as the seedlings grow. In sealed BRIC-PDFU the minimum embryo-surface pO_2 is 13.53 kPa against 13.77 kPa on the ground, and 21% of embryo cells are hypoxic ($pO_2 < 1$ kPa) at 24 h against 20%. With the proxy tissue diffusivity and respiration rate, the stele core of the axis is anoxic in every environment, so the environment changes the size of the hypoxic region rather than whether there is one. The resistance that matters is the seed's own coat and tissue. Sealed hardware accumulates seedling ethylene to a peak of 86.7 ppb in BRIC-PDFU, below the lowest concentration

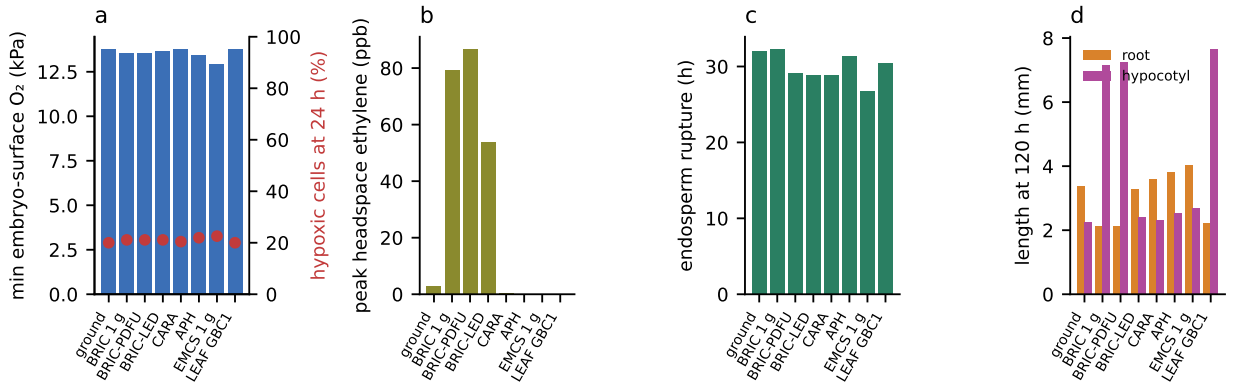


Figure 3: Hardware environments. (a) Minimum embryo-surface O_2 (bars) and percentage of hypoxic embryo cells at 24 h (points). (b) Peak headspace ethylene. (c) Endosperm rupture time. (d) Root and hypocotyl length at 120 h.

(0.125 ppm) that promoted germination of dormant Col-0 seeds [20]; the twin therefore predicts no ethylene triple response from seedling emission alone, consistent with the curated BRIC-PDFU record, which notes that Millar et al. [30] saw no ethylene triple response in sealed PDFUs compared with open dishes. A cabin-air exchange at the Mir-measured 1.1–1.7 ppm would change this: the twin's dark hypocotyl at 10 ppm reaches 812 μm against 7517 μm in air. At the GBC1 seed bed the pump-driven air speed is 0.053–0.116 mm s^{-1} (0.25–0.85 L min^{-1}), giving Sh ≈2.02: the pump renews the headspace but does not ventilate the seed boundary layer.

3.4 Gravity, atmosphere and radiation

On the EMCS centrifuge preset, root tips turn toward gravity at a rate scaled by the gravity–auxin-asymmetry relation and do not bend at 0 g (Fig. 4a). The exploration atmosphere (56.5 kPa, 34% O_2) gives a pO_2 about 10% below sea level and raises gas diffusivity. Because the resistance lies in water and tissue rather than air, faster gas diffusion barely helps: embryo-surface O_2 is 11.93 kPa (sea level 13.77 kPa) and endosperm rupture moves from 32.0 to 33.5 h. Uncompensated air at 56.5 kPa lowers embryo-surface O_2 to 6.97 kPa and delays rupture to 49.9 h; 10% and 5% O_2 at 1 atm delay it to 41.7 and 63.7 h, consistent with the O_2 requirement of germinating Brassicaceae seeds [31]. Radiation has, by construction, no effect unless the hypothesis couplings are switched on (Fig. 4c): with all three set to 1, the solar-particle-event scenario shortens the 120 h root from 3362 to 2795 μm. Seeds at 0 g rupture the endosperm earlier than at 1 g in the same hardware (BRIC-PDFU 29.1 vs 32.3 h; EMCS 24.1 vs 26.8 h) because the *assumed* larger wetted fraction at 0 g speeds imbibition; the size of this effect rests on assumed values.

3.5 Simulated exposure and observed cell-type signatures

At 24 h in sealed BRIC-PDFU, the simulated O_2 deficit of each atlas cell type and the observed low-oxygen enrichment of that cell type in the decoder have Spearman $\rho = 0.77$ ($p = 0.01$, $n = 10$ clusters; Fig. 5). This is a weak test — the decoder scores marker programmes, not exposure — but it is the comparison the two tools are designed to support. The ranking of simulated deficits is the same in every scenario (it is set by the radial geometry), so the agreement tests the model's radial structure, not its response to the environment.

4 Discussion

The twin turns the Bassel et al. embryo-growth mechanism into a seed in an environment. Three predictions follow that flight data can test. First, embryo hypoxia in flight hardware is set by

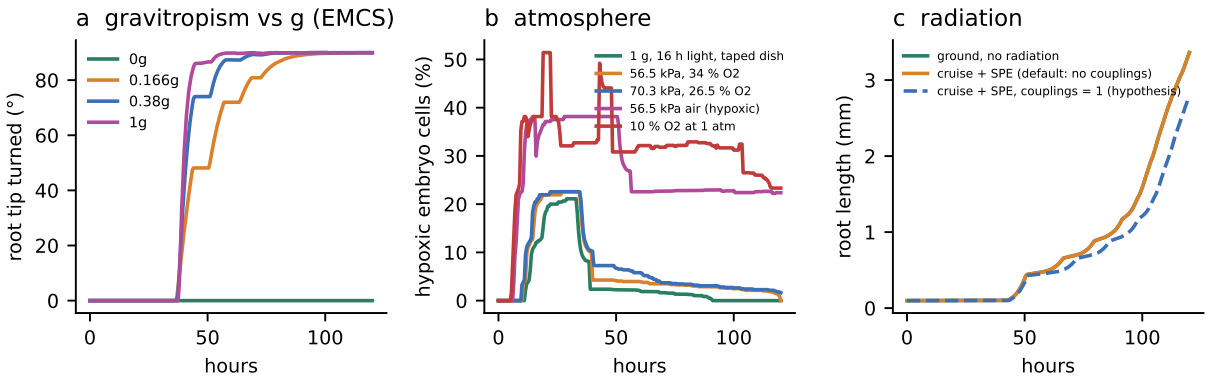


Figure 4: (a) Root tip angle on the EMCS preset at four gravity levels. (b) Percentage of hypoxic embryo cells under exploration and hypoxic atmospheres. (c) Root length under a solar particle event without (solid) and with (dashed) the radiation hypothesis couplings.

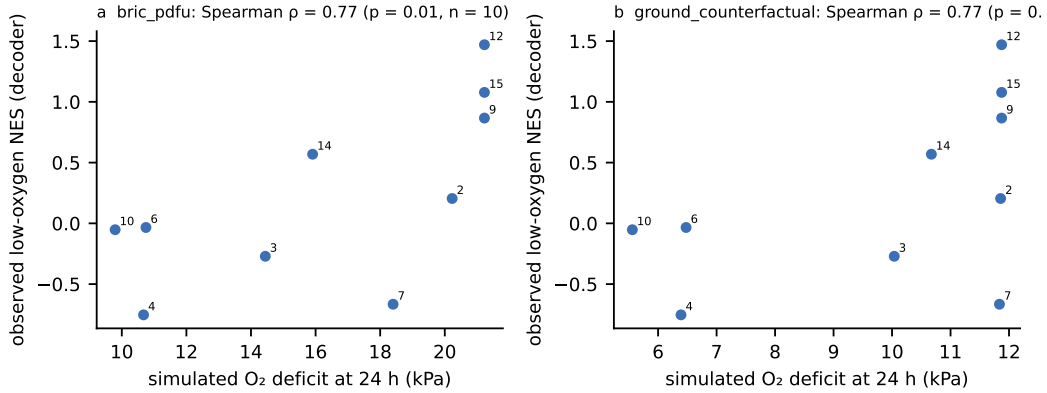


Figure 5: Simulated O_2 deficit at 24 h per atlas cell type against the observed low-oxygen NES of that cell type in the decoder.

the seed coat and tissue, so it should differ little between sealed and ventilated hardware during germination. Second, at seed scale gravity acts through water films and, after protrusion, tropisms — not through the air boundary layer, which is at the still-air limit in sealed dishes and barely ventilated even in the pumped GBC1. Third, seedling ethylene alone does not reach response levels in sealed dishes; cabin ethylene does.

Limitations. The geometry is schematic. 86 parameters remain assumed. Several sources are proxies: *Brassica napus* tissue diffusivity, *Lepidium* cap mechanics, and tomato and apple ACC oxidase. The calibration is not identifiable. The Topham et al. and Bassel et al. supplementary equations could not be retrieved, and the hormone core uses the fully published Abley et al. model. Radiation-to-hormone couplings are hypotheses.

Data and code availability. Code (MIT), ledger, presets, calibration targets, results tables and figures: `https://github.com/dr-richard-barker/deepspace-seed-twin`; live tool: `https://dr-richard-barker.github.io/deepspace-seed-twin/`. Zenodo archive: [TODO: DOI after deposit].

Author contributions, funding, competing interests. [TODO: to be provided by the author]

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